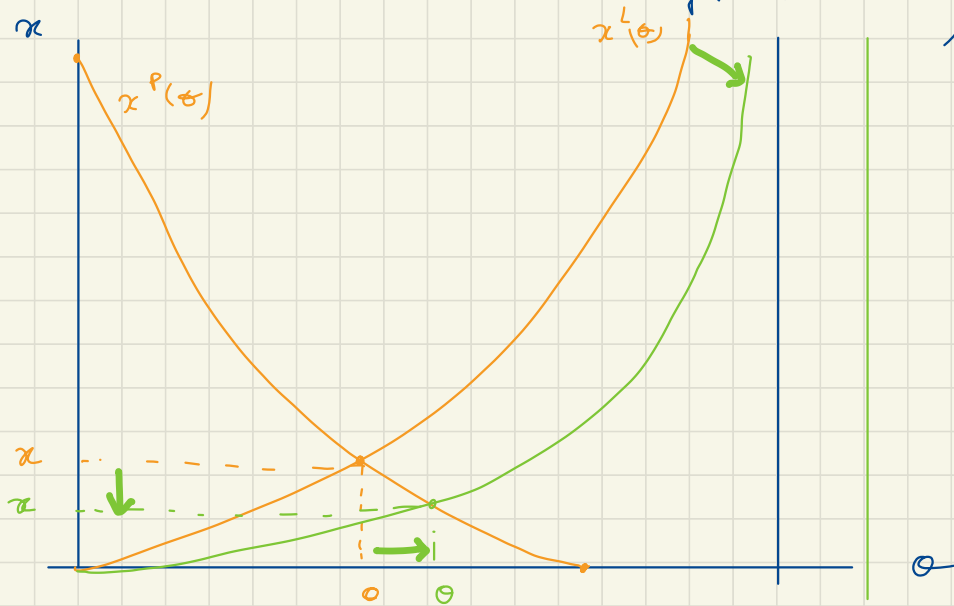


Technology Shocks with Fixed Prices

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Positive technology shock. increase in a

$$\begin{cases} x = x^L(\theta) = f^{-1} \left(\frac{w/p}{a} l^{1-\alpha} \hat{f}(\theta)^{1-\alpha} [1+\hat{\tau}(\theta)]^\alpha \right) \\ x = x^R(\theta) = \tau^{-1} \left(\left[\frac{x^\varepsilon \mu \alpha}{w \cdot l} \quad \frac{1}{\hat{f}(\theta)} \right]^{1/\varepsilon-1} - 1 \right) \end{cases}$$



After an increase in technology a

- $\theta \uparrow$ [$\hat{f}(\theta) \uparrow$, $1-\hat{f}(\theta) \downarrow$, $q(\theta) \downarrow$, $\hat{\tau}(\theta)$]
- $x \downarrow$ [$f(x) \downarrow$, $1-f(x) \uparrow$, $q(x) \uparrow$, $\tau(x)$]
- $l = \hat{f}(\theta) l \propto \underline{l} \uparrow$
- $y = \frac{x^\varepsilon}{[1+\hat{\tau}(x)]^{\varepsilon-1}} \cdot \frac{w}{p} \propto \underline{y} \uparrow$

key difference b/w technology & AD shocks is

the response of $x \rightarrow$ then to look at
product market tightness to separate AD &
technology shocks.

$$- \quad c = \overset{\uparrow}{\gamma} / (1 + \overset{\uparrow}{z}(\overset{\uparrow}{x})) \quad \text{so } c \uparrow$$

$$- \quad \hat{v} = \overset{\uparrow}{l} / \overset{\downarrow}{q(\hat{x})} \quad \text{so } \hat{v} \uparrow$$